Training Mechanics & Evaluation.

QLS–MiCM Workshop

Why Evaluation Matters

Medical ML adds challenges:

- small datasets.
- class imbalance.
- noisy predictors.
- high stakes (false negatives matter).

Accuracy alone is misleading..

We need richer, more reliable evaluation methods.

Cross-Validation for Model Selection

GridSearchCV:

- splits training data into folds.
- trains several models with different hyperparameters.
- selects the model with the best mean CV performance.
- avoids tuning on the test set

Core principle:.

Hyperparameters must be chosen using only training data.

What We Tune in This Exercise

🔧 Architecture.

- `hidden_layer_sizes` : (32,), (64,), (32,16)

🔧 Regularization.

- `alpha` → L2 penalty controlling model complexity

🔧 Training Settings.

- `max_iter` .
- `random_state` (reproducibility).
- optional: `early_stopping=True` to reduce overfitting + speed up training

The goal is to observe **how each change affects test performance**.

Evaluation Metrics

From `classification_report`:

- **Accuracy**.
  .- overall correctness; can be misleading with imbalance.
- **Precision**.
  .- among predicted positives, how many are correct?.
- **Recall (Sensitivity)**.
  .- of true positives, how many were detected?.
- **F1-score**.
  .- harmonic mean of precision + recall; balances both.

Medical importance:

Low recall = missed disease cases.

ROC AUC

If the model supports probabilities:

- ROC AUC measures ranking quality.
- threshold-independent.
- widely used in medical risk assessment.

High AUC means:

Model gives higher scores to true positives more often.

Practical Training Notes

- Use `stratify=y` in train/test split to preserve class ratios.
- Use `n_jobs=-1` to speed up grid search.
- Watch for `ConvergenceWarning` -> increase `max_iter`.
- Don’t build huge grids — slow and prone to overfitting CV folds.